

Teaching

Teaching is one of my most important responsibilities as a faculty member. My goal is not only to convey core biological principles to my students, but to help them develop confidence, intellectual independence, and an appreciation for how scientific discovery unfolds. Because my research centers on how microbial metabolites shape neural function and behavior, I am especially motivated to help students see biology as an integrated discipline—one that connects genetics, neuroscience, microbiology, and behavior into a coherent framework.

My primary undergraduate teaching role has been as course director of **Genetics and Development (BIOL 1030)**, one of four foundational lecture courses for all bioscience majors at Yale. With enrollments approaching nearly 200 students per semester, this course serves students with diverse backgrounds and career goals. Many students in this class have not taken any coursework in genetics for several years. I structure the course to emphasize conceptual understanding over memorization, building complex ideas stepwise from genetic mechanisms to developmental systems. In a class of this scale, fostering engagement is both essential and challenging, so I prioritize clarity of organization with frequent encouragement of questions, and deliberate open-ended questions as part of learning. Students are pushed to participate during “think, pair, share” activities. Creating this atmosphere reduces intimidation and reinforces that science is iterative and collaborative. Watching students grow in their abilities in this course has been one of my great joys as a faculty member.

I also teach **Sensory Neuroscience Through Illusions (MCDB 3290/NSCI 3290)**, a smaller course that uses examples of perceptual phenomena to illuminate neural circuit function. This setting allows deeper interaction and exploration of how experimental approaches reveal principles of sensory processing. The course naturally bridges my research interests in neuromodulation and behavior with broader neuroscience concepts, helping students see how experimental design shapes discovery.

At all levels, I incorporate elements of my own research program to reinforce conceptual understanding and provide students with direct exposure to real data. For example, I teach about neuromodulatory microbes in the graduate **Biology of Bacterial Pathogens II (MBIO 6850)** course and lead a behavioral analysis module in the **Interdepartmental Neuroscience Program Coding Bootcamp**, where I include datasets from my own work. Across all settings, I emphasize three themes: rigorous experimental logic, quantitative reasoning, and clear scientific communication and aim to create an environment where students feel intellectually challenged but fully supported.

Advising

Advising is central to the academic experience, particularly in a research-intensive environment. I formally advise undergraduate majors in MCDB and Neuroscience and regularly serve on doctoral candidacy and thesis advisory committees across MCDB, INP, Microbiology, and related programs. In advising meetings, I focus on helping students clarify their goals while recognizing that those goals often evolve. For undergraduates, discussions may center on course selection, research opportunities, or preparation for medical or graduate school. For graduate students, conversations often involve research direction, committee dynamics, authorship expectations, and long-term career planning. My approach is to listen first, ask probing questions second, and offer guidance that promotes independent decision-making rather than prescriptive solutions. I view advising as part of a broader commitment to fostering an inclusive academic culture in which all trainees can thrive.

Mentoring

Mentoring trainees in my laboratory is possibly the most meaningful aspect of my academic career. My lab includes undergraduate researchers, graduate students, postdoctoral fellows, and postgraduates from diverse backgrounds. Because my research spans microbiology, neuroscience, and genetics, trainees develop interdisciplinary fluency and quantitative skills that position them for diverse career paths. I tailor my mentorship style to experience level, scientific goals, and working style, recognizing that effective mentorship evolves over time. At the core of my mentoring philosophy is the belief that rigorous science flourishes in an environment of trust, shared purpose, and open communication. We meet individually each week to discuss data, troubleshoot experiments, and refine hypotheses. In addition, I hold annual individual development plan meetings focused specifically on long-term scientific and professional goals. I begin each academic year with a “state of the lab” discussion outlining our scientific vision, funding landscape, and expectations, and I solicit open feedback to maintain transparency.

I also prioritize peer learning and collective responsibility. Lab meetings emphasize constructive critique, clarity of presentation, and experimental logic. I encourage trainees to present preliminary data early and often, normalizing iteration and revision. This culture has supported trainees in earning competitive awards, presenting at major conferences, and advancing to medical school, PhD programs, and postdoctoral or academic positions. At the same time, I emphasize that success in science is not measured solely by publications, but by growth in independence, resilience, and intellectual curiosity.

I consider it a privilege to contribute to the scientific and professional development of students at every stage of training, and I remain committed to continually refining my approach as both educator and mentor.